Allamaprabhu Ani

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SUMMARY

Computational mechanics PhD candidate and lead developer of PhAST, an open, matrix-free, differentiable PyTorch solver for phase-field fracture. Lead author of the 2026 solver preprint and the 2026 *Engineering Fracture Mechanics* review of machine learning for fracture. Earlier work delivered aerospace hand-calculation automation in Excel-VBA.

SKILLS

Scientific ML: Python, PyTorch, autograd, CUDA, NumPy/SciPy, scikit-learn, neural operators, PINNs, differentiable simulation, inverse problems.
Numerical engineering: finite elements, phase-field fracture, sparse linear algebra, Akantu, FEniCSx, COMSOL, MSC Nastran/Patran, Abaqus, MATLAB.
Infrastructure: Slurm, Linux, Git, GitHub Actions, Sphinx, Bash.

EXPERIENCE

PhD Student, Computational Mechanics and ML — Jul 2023 - present (writing-up status)
City St George's, University of London

- Lead development and maintenance of PhAST, a matrix-free, GPU-accelerated PyTorch solver for quasi-static and dynamic phase-field fracture.
- Implemented autograd-compatible scalar fracture-toughness recovery and verified gradients against finite differences; extending the method to spatial inverse problems in current thesis work.
- Published six benchmark configurations and an approximately one-million-node NVIDIA A100 study with reproducible reference comparisons.
- Maintained reproducible Slurm pipelines on Hyperion2; local Mac CPU and WSL CUDA development.

Visiting Researcher, Computational Solid Mechanics — 2025
LSMS Lab, EPFL (host: Prof. J.-F. Molinari)

- Lead author of the 2026 *Engineering Fracture Mechanics* review of ML for computational fracture, with collaborators at EPFL and the University of Florida.

Founder and Tool Developer — 2020 - 2021
Aeroknacks India Ltd

- Shipped aerospace hand-calculation automation tools for Excel-VBA and structural-analysis workflows. Built from E.F. Bruhn, Michael Niu, Boeing Design Manuals, and ESDU references; included Cozzone, lug strength, fastener load transfer, buckling, modal, and ABD tools.

EDUCATION

PhD, Mechanical Engineering, City St George's, University of London (Jul 2023 - present; writing-up status). Thesis: matrix-free differentiable fracture mechanics and inverse analysis in PyTorch.
MTech, Design and Manufacturing, National Institute of Technology Silchar (2021 - 2023). CGPA 9.88/10. Best Student Award.
BE, Mechanical Engineering, GM Institute of Technology, VTU (2016 - 2020). CGPA 8.47/10.

SELECTED PUBLICATIONS

Ani, A.S., Molinari, J.-F., Subhash, G., Ponnusami, S.A. (2026). A matrix-free, differentiable PyTorch solver for phase-field fracture: Formulation, benchmarks, and inverse analysis. arXiv:2606.23458.
Ani, A.S., Nakka, R., Subhash, G., Molinari, J.-F., Ponnusami, S.A. (2026). Machine learning for computational fracture and damage mechanics: Status and perspectives. *Engineering Fracture Mechanics* 332, 111778.

AWARDS

Yeoman and Travelling Scholarship (2026); Dean's Award for Outstanding Teaching Support (2026); AFHEA (2024); Funded PhD Studentship (2023); Best Student Award (2023); Royal Aeronautical Society Member (2026).